## Power: USB-C, charger, 3V3, rails

File: power.kicad_sch

## ESP32-S3 module

File: mcu.kicad_sch

## AS5311 magnetic encoder

File: sensor_mag.kicad_sch

## Capacitive front-end (experimental)

File: sensor_cap.kicad_sch

## Display, buttons, LED, piezo

File: ui.kicad_sch

## Accelerometer

File: imu.kicad_sch

PrettyCool Caliper rev A — generated by caliper/hardware/design/build.py from caliper/hardware/design/sheets/*.py.
Do not hand-edit the .kicad_sch files: change the sheet modules and rebuild.
Architecture of record: caliper/hardware/design/SPEC.md

**PrettyCool Boards**

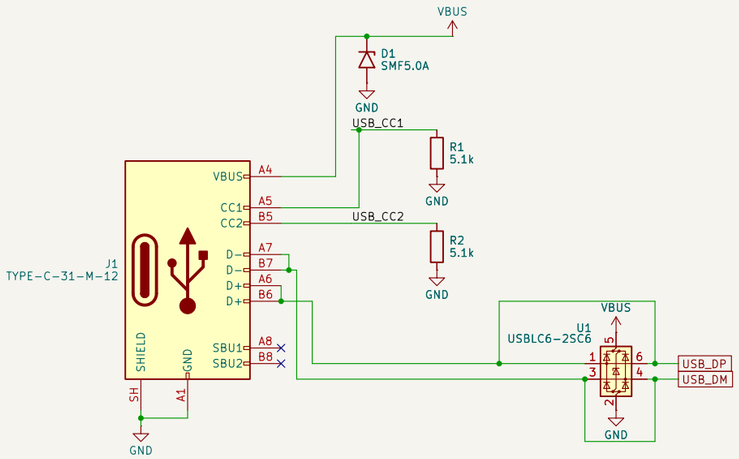
Sheet: /
File: prettycool-caliper-slider.kicad_sch

***Title: PrettyCool Caliper slider board***

| Size: A3 | Date: 2026-11-07 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 1/7 |

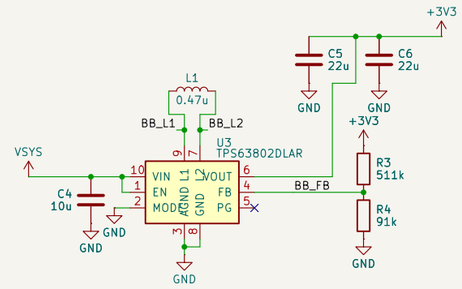
## USB-C input and protection

*HRO TYPE-C-31-M-12 (USB 2.0 device, sink only). Separate 5.1k Rd per CC pin; D+/D- pairs tied at the connector. USBLC6-2SC6 + SMF5.0A within 5 mm of J1, D+/D- routed through the ESD pads as a 90R pair. No series resistors on D+/D- (inside the ESP32-S3, SPEC §5). Right board edge, y 26.5 (SPEC §2.1).*



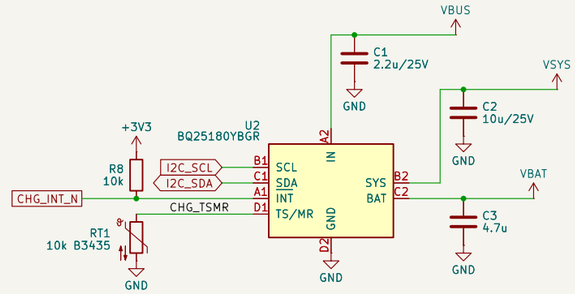

## 3V3 buck-boost: TPS63802 (VSYS 2.5-4.6 V -> +3V3, 2 A)

*Buck-boost, not an LDO: Wi-Fi TX peaks ~355 mA (module datasheet) and the cell falls below 3.4 V (SPEC §3). EN tied to VSYS (always on), MODE = GND (power save: 11 uA Iq), PG unused. R3/R4 = 511k/91k -> 3.31 V (VFB 0.5 V, TI Table 10-5); the divider draws 5.5 uA (R2 <= 100k rule). L1 DFE201612E-R47M (TI Table 10-2). CIN 10u + COUT 22u per Fig. 10-1, plus 22u bulk C6 for the RF bursts. Layout: VIN/VOUT caps and L1 on the same layer as U3, GND pad 8 to the plane with >= 2 vias.*



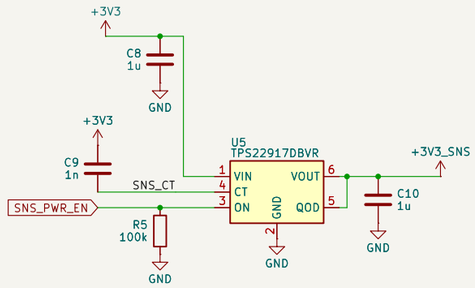

## Charger and power path: BQ25180 (I2C 0x6A)

*ILIM / ICHG (250 mA) / SYS_REG / ship mode set over I2C: register init per dial-me-spinner docs/firmware.md §13.1. IN 2.2u/25V X7R 0805, SYS 10u/25V, BAT 4.7u: each within 1 mm of its ball and of the D2 GND ball. Ship mode exits only on a USB plug-in (TS/MR is not wired to a button in rev A). /INT -> CHG_INT_N (10k pull-up R8, GPIO3). RT1: 10k NTC next to J2 (board temperature, not the cell).*



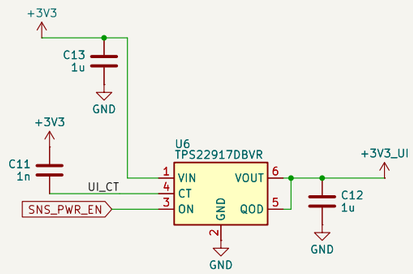

## Sensor rail +3V3_SNS: TPS22917 (SNS_PWR_EN)

*AS5311 (~16 mA) + capacitive front-end. Off in sleep. CT 1n to VIN (slew ~ms, no +3V3 droop): QOD tied to VOUT (internal R) so the rail discharges when off. R5 holds ON low while the ESP32 is in reset. Firmware: drive MAG_CSN / MAG_CLK low before SNS_PWR_EN goes low (no back-powering through the ESD diodes).*



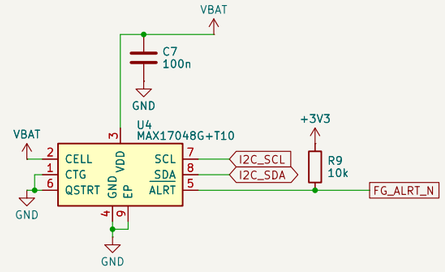

## UI rail +3V3_UI: TPS22917 (also on SNS_PWR_EN)

*LCD logic VDD (ST7789 sleep-in still draws 15+5 uA typ, DM-TFT114 §5.3) + WS2812 RGB LED: both off in deep sleep. Shares SNS_PWR_EN (no free GPIO): the rails are on together whenever the caliper is awake. C13 1u at VIN (pin 1): the switch's own input cap (C5 is 12.5 mm away by copper). Firmware: drive the LCD_* lines and RGB_DIN low before switching off.*



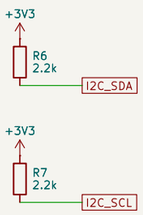

## Fuel gauge: MAX17048 (I2C 0x36)

*Powered from the cell (VDD = CELL = VBAT, Fig. 14). CTG, QSTRT and EP to GND. 3 uA in hibernate. ALRT -> FG_ALRT_N (10k pull-up R9, GPIO2: low SOC / voltage alert).*



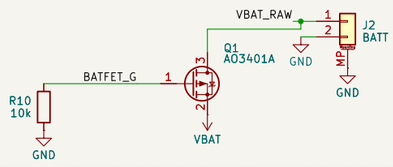

## I2C + open-drain pull-ups (always-on +3V3)

*SDA/SCL 2.2k: BQ25180 0x6A, MAX17048 0x36, LIS2DW12 0x19 at 400 kHz (SPEC §6): ~205 ns rise (30-70 %) with ~110 pF (4.7k gave 438 ns > 300 ns); 1.3 mA sink at VOL 0.4 V (fast mode allows 3 mA). Must stay on the always-on rail: the MAX17048 reads a low bus as 'disconnected' and sleeps.*




## LiPo connector J2 (402535, ~500 mAh, JST-SH 2-pin) + reverse-polarity P-FET Q1

*Pin 1 = BAT+, pin 2 = GND. Pack leads are not standardised: check every pack's polarity anyway. Q1 AO3401A in the BAT+ lead: drain to J2, source to VBAT, gate to GND via R10. The body diode conducts for a correct cell, then VGS = -VBAT turns the channel on (both directions, so charging works). A reversed cell leaves VGS = 0: off. VGS max +-12 V vs 4.4 V: no zener. 0.5 A: <= 43 mV / 21 mW (85 mOhm max at VGS -2.5 V). The cell sits in the housing pocket behind the board (SPEC §2.1).*



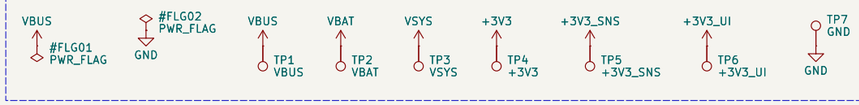

## Mounting holes and fiducials

*H1 (18.75, 24) support hole; H2-H5 tab clamp holes (-3.5, 5.5) (-3.5, 14) (65.5, 5.5) (65.5, 14), plated, to GND (SPEC §2.1). H2-H5 carry an SMT M2 standoff, fitted by hand (no LCSC part fits the 6 mm tab). 3 fiducials.*




## Power flags and test points

*VBUS (USB-C) and GND are driven from outside the board.*




**PrettyCool Boards**

Sheet: /Power: USB-C, charger, 3V3, rails/
File: power.kicad_sch

**Title: Power: USB-C, charger, 3V3, rails**

| Size: A3 | Date: 2026-11-07 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 2/7 |

## ESP32-S3-MINI-1-N4R2 (4 MB flash, 2 MB PSRAM, PCB antenna)

*Pin map = nets.MCU_PINS (SPEC §5); caliper/hardware/design/check_contract.py checks the netlist against it.*
*Antenna end on the top board edge (y = 34), hanging clear of the steel beam; copper keep-out on all layers under the antenna (Espressif HW design guidelines). EPAD + GND pads to the plane with a via array.*
*N8R2 is not stocked at LCSC (2026-11); N4R2 C3013941 is. IO26 is the PSRAM CS: left unconnected.*

## 3V3 decoupling at the module

*Datasheet Fig. 9-1: 22u + 100n at pad 3. C21 100n + C23 10u (0402, 0.5 mm tall) on the BOTTOM, under pad 3, fed from its escape via, own GND vias (the housing floor is 0.75 mm below there, SPEC §2.1).*
*C20 22u (0603, 0.8 mm) is the bulk on top below the module, fed through the In2 +3V3 pour.*

## EN reset delay

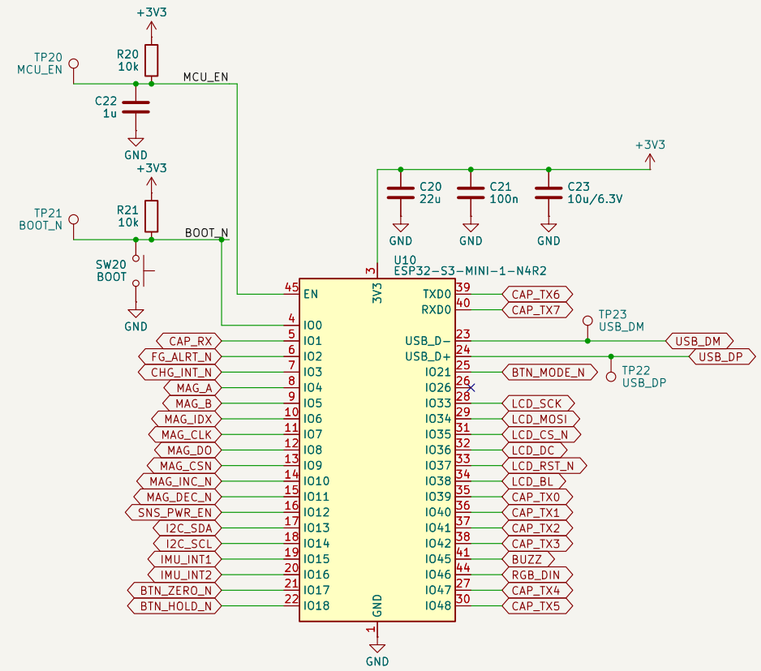
*R 10k / C 1u (datasheet §9). EN pad on a test point for a manual reset.*

## BOOT (GPIO0 strap)

*10k pull-up: SPI boot. Hold SW20 during reset/USB plug-in for the ROM download mode.*
*Normally flashed over USB-Serial-JTAG without the button; SW20 is the recovery path.*

## Test points

*EN, BOOT and the native USB pair for bring-up*




## Strapping pull-downs

*GPIO45 = VDD_SPI select, must read 0 at reset (3.3 V flash): 100k on the BUZZ gate is on the ui sheet (R56).*
*GPIO46 must be 0/float at reset: 100k on RGB_DIN is on the ui sheet (R55).*
*GPIO3 (CHG_INT_N, 10k pull-up R8): JTAG-source strap, ignored while the JTAG_SEL eFuse is unburnt.*

**PrettyCool Boards**

Sheet: /ESP32-S3 module/
File: mcu.kicad_sch

***Title: ESP32-S3 module***

| Size: A3 | Date: 2026-11-07 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 3/7 |

## AS5311 linear encoder (BOTTOM side, over the magnetic tape)

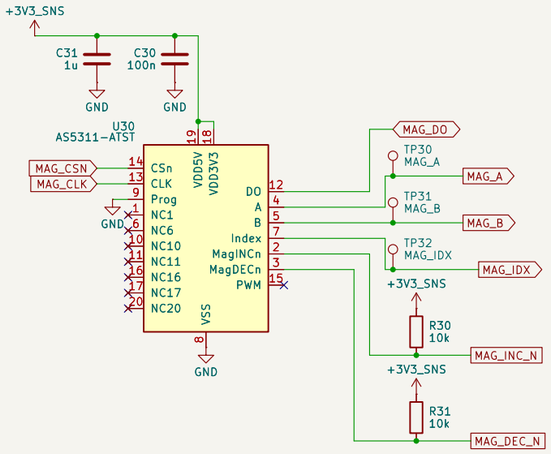
*Centred at (31, 6.0) on the bottom, long axis along X (beam axis), SPEC §2.1. Air gap 0.3 +/−0.1 mm to the tape.*
*3.3 V operation: VDD5V and VDD3V3 both on +3V3_SNS (datasheet §7.7), 100n + 1u at pin 19.*
*A/B −> PCNT (4x: 0.488 um/count), Index once per 2 mm pole pair, SSI (CSn/CLK/DO) for the 12−bit absolute position + OCF/COF/LIN/MagINC/MagDEC. CSn must be low for A/B/Index. Prog to VSS (no OTP in rev A). PWM unused.*
*No copper pour, vias or magnetic parts in the tape window under the IC beyond what the bottom layer rules allow.*

## Magnetic−field indicator pull−ups

*MagINCn / MagDECn are open drain (datasheet Table 1, §4.1): 10k to +3V3_SNS, so they read low (= 'alarm') when the sensor rail is off; the firmware only reads them with SNS_PWR_EN high.*

## Test points

*Quadrature + index for a scope / logic analyser during bring−up*




**PrettyCool Boards**

Sheet: /AS5311 magnetic encoder/
File: sensor_mag.kicad_sch

***Title: AS5311 magnetic encoder***

| Size: A3 | Date: 2026−11−07 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 4/7 |

## TX phase isolation jumpers (0R, DNP to disconnect the experiment)

*Fit R60–R67 to drive the electrodes; leave them off and the eight GPIOs are free test pads.*

## TX series resistors (100R: edge-rate limit, EMI, GPIO protection)

*Next to the comb, on the electrode side of the jumpers. Phase n drives every 8th finger (pitch 0.635 mm).*

## Electrodes (bottom copper, E40 = seed pads; not a BOM part)

*Comb x = 11…51, y = 12.5…17.5; RX strip y = 10.6…11.8 (SPEC §2.1). Bottom mask open over both.*
*Guard GND around the RX strip and the CAP_EL_RX trace on all layers; nothing else on the bottom in y = 9…19.*

## RX jumper + charge amplifier (U40A)

*Q = C*V: Cf 10p C0G || Rf 10M (HPF corner 1.6 kHz, far below the 25–100 kHz drive). Output centred on VREF.*
*Keep the CAP_RX_IN node tiny, guarded, no mask opening on it except at the electrode.*

## VREF = +3V3_SNS / 2, buffered (U40B)

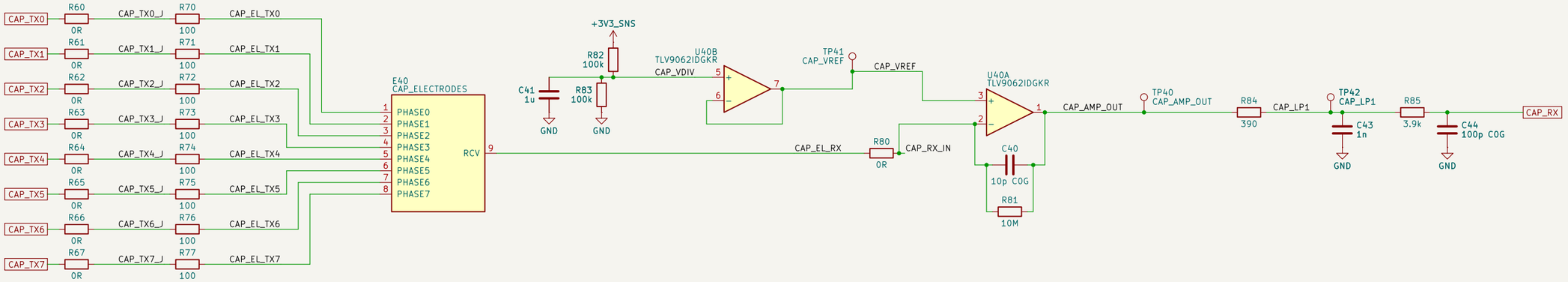
*100k/100k divider (16.5 uA, only while +3V3_SNS is on) filtered by C41 1u, unity-gain buffer U40B.*
*The buffer output drives only U40A's + input (no capacitive load on the op-amp output).*

## 2nd-order RC low-pass (~408 kHz per pole) -> CAP_RX (ADC1_CH0)

*390R/1n then 3.9k/100p C0G (second stage 10x the impedance, so it barely loads the first).*
*Anti-alias + kills the op-amp's HF noise; the synchronous demodulation is done in firmware.*
*Layout: R84/C43 at U40, R85/C44 at the ESP32's GPIO1 pad (bottom, under the module): the long run*
*is CAP_LP1 (390R + 1n at the source end) and CAP_RX is a few mm from the ADC pin.*

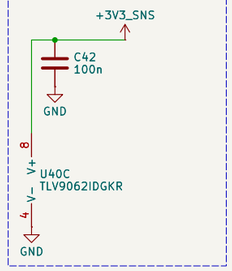
## Test points

*Charge-amp output, VREF and the filter's first stage (CAP_LP1) for scope probing.*
*TP42 is on CAP_LP1, not CAP_RX: CAP_RX ends at the ESP32 (R85/C44 there), and a pad in the*
*test-pad row would hang a 30 mm stub on the ADC node.*




## Op-amp supply

*100n at pin 8*



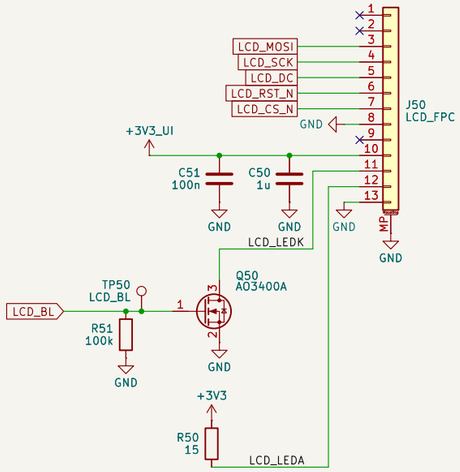

| | | |
|---|---|---|
| **PrettyCool Boards** | | |
| Sheet: /Capacitive front-end (experimental)/ | | |
| File: sensor_cap.kicad_sch | | |
| ***Title: Capacitive front-end (experimental)*** | | |
| Size: A3 | Date: 2026-11-07 | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 5/7 |

## LCD: 1.14in 240x135 ST7789 IPS, 13-pin 0.5 mm FPC (J50)

*Pin map: DM-TFT114-397 §3.1 (1 NC, 2 NC, 3 SDA, 4 SCL, 5 RS=D/C, 6 RESET, 7 CS, 8 GND, 9 NC, 10 VDD, 11 LEDK, 12 LEDA, 13 GND).*
*SPI2 40 MHz write-only (SPEC §6). VDD from the switched +3V3_UI rail. 1u + 100n at pin 10.*
*Beside the module on top; the glass is glued to the bezel, not the board (SPEC §2.1).*
*Check the bought panel's pin order and 0.5 mm tail pitch before ordering (vendors differ).*

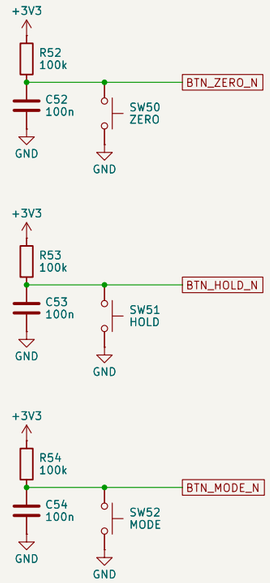
### Backlight: low-side N-MOSFET on LCD_BL (PWM)

*+3V3 -> R50 15R -> LEDA, LEDK -> Q50 -> GND. 1 white LED, VF 3.0 V (2.9-3.1) at 20 mA.*
*(3.31 V - VF) / 15R: 20 mA at VF 3.0, 27 mA max at 2.9, 14 mA at 3.1 (10R gave 31-41 mA).*
*Headroom is only ~0.3 V, so brightness follows VF and the +3V3 tolerance; dim by PWM (LEDC, ~20 kHz).*
*R51 100k holds the gate low while GPIO38 floats in reset.*



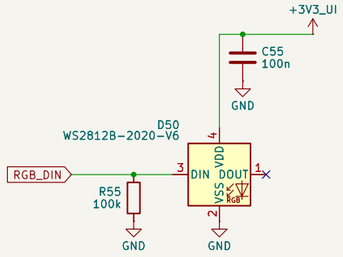

## Buttons: ZERO / HOLD-SEND / MODE (active low, RTC GPIOs 17/18/21 for deep-sleep wake)

*100k pull-up (no static current) + 100n to GND: 10 ms RC debounce; ext1 wake on any-low.*
*TS-1088 3.9x2.9 mm top tact switches on the top edge band: ZERO x-8, HOLD x-20, MODE x-54 (SPEC §2.1).*



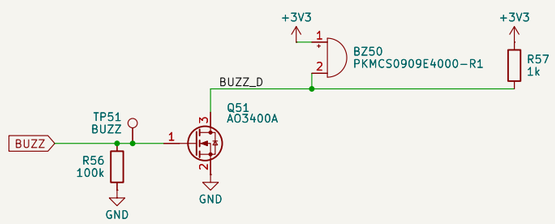

## RGB status LED: WS2812B-2020-V6 on +3V3_UI (GPIO46 = RGB_DIN)

*3.3-5.3 V part (LCSC C52917434), so no level shifter: DIN at 3.3 V logic = VDD. Off with +3V3_UI in sleep.*
*R55 100k pull-down: GPIO46 is a strap (must be 0/float at reset). 100n at VDD.*




## Piezo: Murata PKMCS0909E4000-R1 (4 kHz, external drive), low-side MOSFET on BUZZ (GPIO45)

*+3V3 -> BZ50 || R57 1k -> Q51 drain. LEDC square wave at ~4 kHz on the gate. R57 recharges the piezo*
*while Q51 is off (3.3 mA only while Q51 conducts). R56 100k gate pull-down: GPIO45 is the VDD_SPI strap*
*and must read 0 at reset (SPEC §5).*




**PrettyCool Boards**

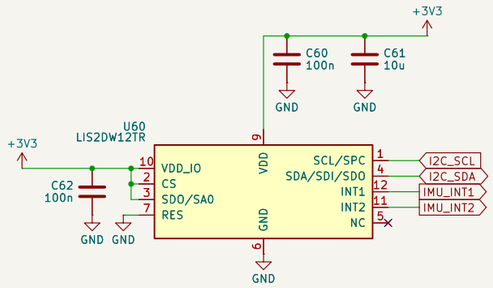
Sheet: /Display, buttons, LED, piezo/
File: ui.kicad_sch

***Title: Display, buttons, LED, piezo***

| Size: A3 | Date: 2026-11-07 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 6/7 |

## Accelerometer: LIS2DW12 on the system I2C (addr 0x19, SA0 high)

*Always on (+3V3): wake-up / 6D / activity interrupts while the ESP32 is in deep sleep (~1 uA in low-power wake mode).*
*CS high = I2C. SA0 high (never ground it: internal pull-up, see module docstring). RES to GND.*
*INT1 -> IMU_INT1 (GPIO15, RTC: deep-sleep wake on pickup), INT2 -> IMU_INT2 (GPIO16). Push-pull, no pull-ups.*
*100n + 10u at VDD (pin 9), 100n at VDD_IO (pin 10). Keep away from the buck-boost inductor.*




**PrettyCool Boards**

Sheet: /Accelerometer/
File: imu.kicad_sch

***Title: Accelerometer***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-11-07 | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 7/7 |